

How Good is Your Heifer Rearing Program?

Dr. Christian Heyerhoff

Traditionally, we measured our heifer rearing programs by monitoring first lactation BCAs, 305MEs and plotting heifer growth curves on established templates. BCAs and 305MEs are not the best measure because they are largely historical in nature, and most of the available heifer growth curve templates are too conservative for the dairy industry in Canada today. What are the best measuring tools to use? They must be timely (how are our heifers performing today), accurate and easy to access.

Is there a good place to start? Recently, I became aware of a new index developed by Dr. Gavin Staley, a researcher with Diamond V who examined the records of thousands of dairy cows from large herds in the U.S. to reach his conclusions. That index was **1st Lactation 10 week milk production** (1st lactation cows tend to peak at around 10 weeks of lactation). What he discovered was: heifer rearing programs were meeting goals/standards if "1st lactation 10 week milk production meets or exceeds the average milk production of the herd". This measure effectively provides a true measure of the adequacy of heifer rearing in one simple step. It is important to note that the farm production records must be on Dairy Comp 305 to calculate this index. I have evaluated this index on several of the dairy farms I service, and have found it to be very useful and informative. If you are meeting this standard, congratulate yourself for doing an outstanding job raising heifers. If not, you will need to determine where the issues lie.

There are critical control points in heifer rearing that we can measure that will allow us to identify where the weaknesses exist. I will present six key monitoring points in heifer rearing that will help you identify any major issues that exist on your farm.

First Step: Evaluate the farms current performance of 1st lactation cows:

(see Appendix I for DC305 item adds and command to replicate the tables below.)

Week 10 Milk by Lactation Group

By LCTGP	Pct	Av MILK	AvW10MK	Av PEAK
-----	----	-----	-----	-----
1	41	38	40.7	42
2	27	41	50.1	52
3	32	42	52.8	55
=====	=====	=====	=====	=====
Total	100	40	47.2	49

Table 1 – Farm A is reaching their goal of 1st lactation week 10 milk production (40 L). The week 10 milk production of 1st lactation cow is directly in-line with the herd average.

Week 10 Milk by Lactation Group

By	LCTGP	Pct	Av MILK	AvW10MK	Av PEAK
	1	36	34	36.0	37
	2	25	38	46.4	49
	3	39	40	49.9	51
Total		100	38	44.1	45

Table 2 - Farm B is not reaching their goal of 1st lactation week 10 milk production (40 L). Farm B's 1st lactation animals are underperforming comparatively to the rest of the herd.

Monitoring Point # 1: 1st Lactation Milk Production by Age at Freshening

Goal: < 2 litre spread in milk production between heifers by age at freshening
(see Graph 1 and 2)

Monitoring Point # 2: Calving Time

Key measurement A: Weight of 1st lactation heifers post-calving.

Goal: $\geq 85\%$ of the mature weight of the herd.

Key measurement B: Stillbirth rate of 1st Lactation.

Goal: 1st lactation stillbirth rate is $\leq 8\%$.

Ensuring ideal weight and frame at calving allows heifers to calve without complications and milk efficiently. There is less demand of energy for growth and more energy for milk production.

If you are not achieving these goals: Examine nutrition, disease rates, size at first breeding calving area comfort, movement and assistance protocols.

NOTE: This assumes that transition health is not a concern. If it is, this needs to be addressed too.

Monitoring Point # 3: Breeding Size

Key Measurement A: Weight at 1st breeding.

Goal: Weight is $\geq 55\%$ (~ 900lbs) of the average mature cow weight.
Achieve this weight by 13 months of age.

Key Measurement B: Withers height at breeding.

Goal: Withers height is ≥ 50 inches at breeding

Key measurement C: Pregnancy Rate.

Goal: Pregnancy rate $\geq 30\%$

We need to change our focus from “**age** at 1st breeding” to “**size** at 1st breeding” so that heifers freshen at the ideal size. A high pregnancy rate demonstrates that heat detection is excellent, the conception rate is high and problem breeders are promptly culled. Heat detection aids such as activity monitors are worth every penny in heifers.

Can we maximize 1st lactation performance while maintaining an early age at first freshening? Yes, based on both research and on evidence from top performing Canadian dairy herds.

Monitoring Point # 4: Six months of Age

Key Measurement: Heifer weight at 6 months of age

Goal: $\geq 30\%$ (~500 lbs) of the average mature cow weight.

The rate of growth in heifers is much faster in the first 6 months than afterwards. If they are not meeting goals at this age, they never reach the later goals because there is no compensatory growth after 6 months of age. If you are not achieving this goal, evaluate nutrition, disease rates and weaning protocols (monitoring points 5 and 6).

Monitoring Point # 5: Time of Weaning

Key Measurement A: Weight at 56 days of Age

Goal: Average daily gain (weight at 56 days of age – birthweight) is ≥ 2 lbs/day.

Key measurement B: Blood ketone (BHB) levels near the conclusion of the weaning process

Goal: 100% of calves have ≥ 0.2 mmol/L the day before intended completion of weaning.

This is most stressful part of a heifer’s life with the possible exception of calving. Measuring BHB levels is a tool that has been validated to use as an indicator of grain intake and rumen development in pre-weaned calves. This period is a common time during which heifer growth is impaired, which affects future performance.

If you are not achieving these goals, evaluate early calf nutrition, disease rates (pneumonia, scours, navel issues), pre-weaning feeding practices and the weaning process.

Monitoring Point #6: First Week of Life

Key Measurement A: Blood total protein (collected from calves 1 to 9 days old).

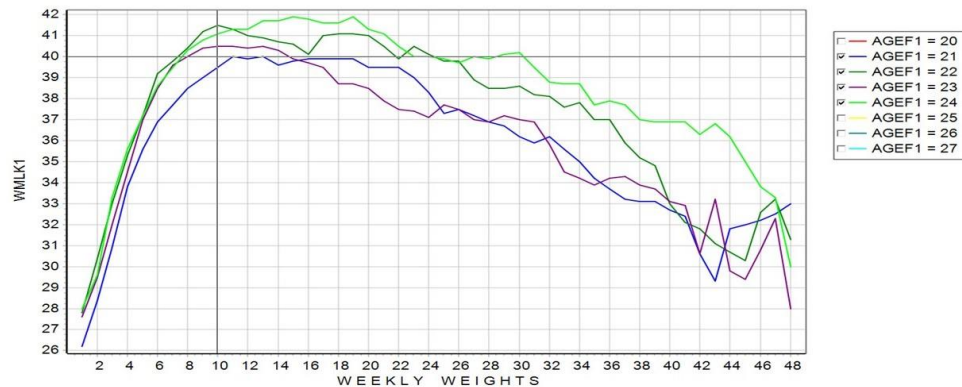
Goal: >85% of calves have total protein greater than 5.5 g/dl.

Key Measurement B: Weight at birth (used to calculate average daily gain)

It is important for calves to get the right start to life. If you are not achieving these goals, evaluate colostrum protocols focusing on the quality and the quantity of the colostrum, timing of colostrum feeding and the cleanliness of the colostrum.

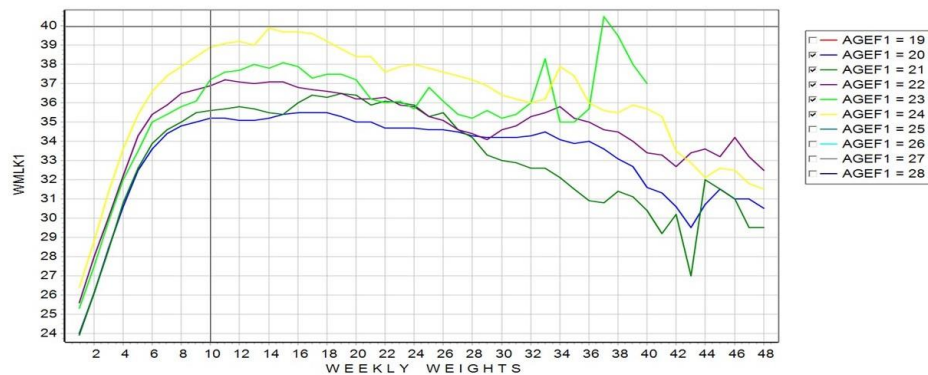
So, what is your 1st lactation week 10 milk production??

1st Lactation Weekly Milk by Age Fresh



Graph 1 - Farm A has minimal spread in milk production between heifers calving between 21 - 24 months. This farm has young, productive animals which are meeting their on-farm goals.

1st Lactation Weekly Milk by Age Fresh



Graph 2 - Farm B has a larger (4 L) spread in milk production for heifers calving in at 21 vs. 24 months. This is not meeting the on-farm goals for milk production. This farm is currently working through monitoring points 2 – 6 to identify areas of opportunity.

Appendix I - Instructions of adding W10MK Item in dairy comp 305 and running the command to bring up average week 10 milk production. Steps: 1. Type Alter in command line 2. Select Items 3. Add item

Item Name	W10MK
Item type	122
Value	10
Description	Average Week 10 Milk
Command Line	SUM MILK W10MK PEAK FOR MILK>0 BY LCTGP